

# **EDGE INTELLIGENCE**

## **LAB ASSESSMENT**

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### **Log Data Analytics**

#### **Objective**

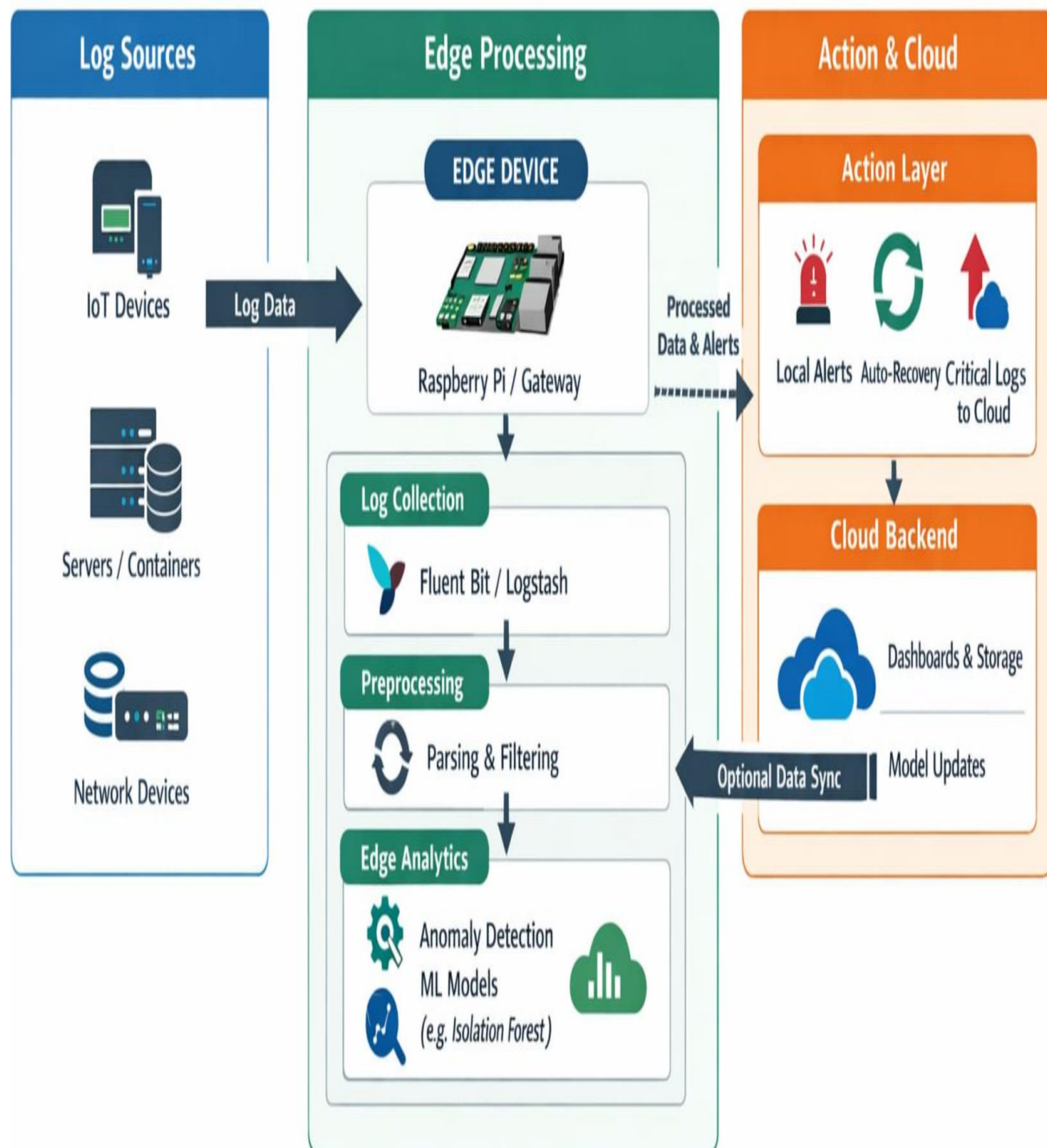
Build a system that performs real-time log data analysis at the edge to:

- Detect anomalies instantly
- Reduce cloud dependency and latency
- Improve system reliability and response time

#### **Use Case**

- Monitoring logs from:
  - IoT devices
  - Servers / containers
  - Network devices
- Example:
  - Detect unusual spikes in errors
  - Identify security threats (e.g., failed login bursts)
  - Predict failures before they occur

## Architecture Overview



### Data Flow:

Log Sources → Edge Device → Local Processing → Alerts → Cloud (optional)

### Components:

- **Log Sources:** Applications, IoT sensors, servers
- **Edge Node:** Raspberry Pi / Gateway / Local server

- **Processing Engine:** Lightweight ML or rule-based system
- **Cloud Backend:** For storage, dashboards, model updates

## **Data Processing Pipeline**

### **Step 1: Log Collection**

- Tools: Fluent Bit, Logstash (lightweight config)
- Collect logs in real-time

### **Step 2: Preprocessing**

- Remove noise
- Parse logs (JSON, text)
- Feature extraction (timestamp, error type, frequency)

### **Step 3: Edge Analytics**

- Rule-based detection (thresholds)
- ML models:
  - Anomaly detection (Isolation Forest)
  - Time-series forecasting (ARIMA / LSTM-lite)

### **Step 4: Action Layer**

- Trigger alerts locally
- Auto-restart services
- Send critical logs to cloud

## **Edge Intelligence Techniques**

- **On-device ML inference** (TinyML / optimized models)
- **Sliding window analysis** for real-time detection
- **Event filtering** to reduce data transfer
- **Federated learning** (optional for model updates)

## **Technology Stack**

- **Languages:** Python, Go
- **Edge Frameworks:** TensorFlow Lite, ONNX Runtime
- **Log Tools:** Fluent Bit, Filebeat
- **Messaging:** MQTT, Kafka (lightweight edge brokers)
- **Cloud:** AWS IoT / Azure IoT / GCP

## **Advantages**

- Low latency (real-time insights)
- Reduced bandwidth usage
- Improved privacy (local processing)
- Faster incident response

## **Challenges**

- Limited compute at edge
- Model optimization required
- Device management complexity
- Security at edge nodes

## **Success Metrics**

- Detection latency (< 1 sec)
- Reduction in cloud data transfer (%)
- Accuracy of anomaly detection
- System uptime improvement

## **Future Enhancements**

- Self-learning models at edge
- Integration with DevOps (CI/CD alerts)
- AI-driven root cause analysis
- Edge + Cloud hybrid intelligence